Proteomics from cells to the clinic

Mass-spectrometry-based proteomics: from single cells to clinical applications

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The review's argument: sensitivity and scale unlock new biology

Why measure proteins directly?

- Abundance depends on translation, turnover and secretion as well as transcription.
- Modification, localization and interaction determine functional state.
- A gene or transcript measurement does not fully specify those properties.

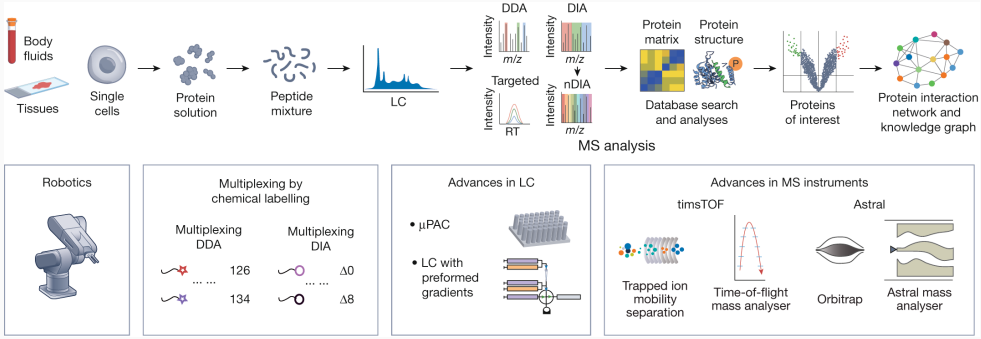
Two expanding frontiers

- Less material: individual cells and selected regions within tissues.
- More samples: automated workflows for patient cohorts and clinical assays.

This is a broad narrative review, not a new assay or a unified benchmark.

Guo et al., pp. 901–902. Claims and performance examples reflect the literature covered in the 2025 review.

A common workflow links diverse applications



Guo et al., Fig. 1 (original). LC: liquid chromatography; MS1: precursor ions; MS2: fragment ions.

What the instrument actually measures

1. Extract proteins and digest them into peptides; optionally enrich a selected class.
2. Separate peptides by LC, then electrospray them into gas-phase ions.
3. Measure precursor mass-to-charge ratios; fragment ions provide sequence evidence.
4. Match spectra to sequences, control identification error, integrate ion signals, and infer proteins.

Running example: comparing two cell states
Multiple peptides support a protein's abundance change. A modified peptide, a cross-linked peptide pair or a changed cleavage site answers a different question and requires a different workflow.

Guo et al., pp. 901–904 and Fig. 1. Running example is a teaching schematic.

Progress comes from the whole measurement chain

Stage	Advance highlighted in review	Practical consequence
Preparation	96-well robotics; bead-based digestion; FFPE compatibility	More reproducible handling; access to archived tissues
Separation	Preformed LC gradients; microflow; micropillar columns	Higher throughput and retention-time robustness
Instrumentation	Ion mobility/PASEF; Orbitrap–Astral	More useful ion sampling; faster, more sensitive measurements
Computation	DIA interpretation; predicted spectral libraries	Extract sequence and quantitative evidence from complex spectra

Guo et al., pp. 902–903. PASEF: parallel accumulation–serial fragmentation; FFPE: formalin-fixed paraffin-embedded.

DDA, DIA and targeted MS optimize different tasks

Mode	Selection rule	Strength	Tradeoff
DDA	Select abundant precursor ions for fragmentation	Relatively simple spectra; useful for affinity-purified or cross-linked material	Stochastic selection yields missing values
DIA	Cycle through mass windows and fragment ions within each	Consistent sampling; fewer missing values	Mixed spectra need computational deconvolution
Targeted	Preselect peptides and fragment signals	Sensitive, precise measurement of a defined panel	Limited discovery outside the panel

For DIA quantification, the review recommends at least four, ideally near ten, measurements across an elution peak.

Guo et al., pp. 902–903, Box 1. DDA: data-dependent acquisition; DIA: data-independent acquisition.

Quantification: sample multiplexing is not free sensitivity

Label-free / DIA

- Compare peptide intensities across runs.
- Consistent acquisition reduces missingness.
- Retention-time alignment, interference correction and run comparability still matter.

Isobaric labels / TMT

- Pool labelled samples; reporter ions encode their relative abundances.
- Co-isolated unrelated peptides can distort abundance ratios.
- Narrow precursor isolation or additional MS3 can reduce interference.

Absolute quantification
Known amounts of stable-isotope-labelled reference peptides provide an internal scale, especially valuable when transferring clinical assays.

Guo et al., p. 903, pp. 907–908 and Fig. 3. TMT: tandem mass tags.

Choose the workflow by the biological quantity

Question	Additional operation	Readout / interpretation
Which proteins change?	Broad expression profiling	Peptide-derived protein abundance
Which partners associate?	Affinity purification or proximity labelling	Enriched partners or molecular neighbourhood
Which sites are regulated?	PTM-specific enrichment	Site-localized modified peptide abundance
What changes structurally?	Exchange, cross-linking or limited proteolysis	Exposure, distance constraints or accessibility
Which targets engage a drug?	Chemical probes or thermal perturbation	Activity-associated labelling or stability response

Synthesis of Guo et al., pp. 903–905. These readouts need distinct controls; they are not interchangeable.

Functional proteomics: separate signal from mechanism

Interactions Affinity purification captures associated complexes; proximity labelling captures a local neighbourhood.

The review highlights a near-complete yeast interactome and integration with AlphaFold-Multimer to interpret interfaces.

Post-translational modifications Enrichment enables measurement of tens of thousands of localized PTMs.

Relative modified-peptide change differs from occupancy: the fraction of a protein population carrying a modification.

Journal club interpretation

Association is not necessarily direct binding. A PTM change may reflect altered protein abundance; occupancy needs modified and unmodified information.

Guo et al., p. 904; cited examples: Michaelis et al. (2023), Prus et al. (2024). Interpretive cautions are ours.

Structural and chemical assays provide complementary evidence

Approach	Measurement principle	What it helps establish
HDX-MS	Exchange of exposed amide hydrogens for deuterium	Changes in protection, conformation or binding interfaces
XL-MS	Covalently link proximal residues and identify paired peptides	Spatial restraints within and between proteins
LiP-MS	Compare accessibility to limited proteolysis	Conformational or ligand-associated accessibility changes
ABPP / TPP	Probe labelling / thermal denaturation response	Enzyme activity profiles / candidate drug targets

Our interpretation: a structural constraint or stability shift narrows hypotheses; orthogonal tests establish the proposed mechanism.

Guo et al., pp. 904–905. ABPP: activity-based protein profiling; TPP: thermal protein profiling.

Single-cell proteomics: protect a scarce, unamplifiable input

A typical cell contains only **50–500 pg protein**.

- No amplification analogous to RNA amplification.
- Miniaturized droplets and microfluidics reduce surface losses.
- Low LC flow and sensitive instruments increase usable signal.
- DIA and multiplexing improve sampling or throughput.

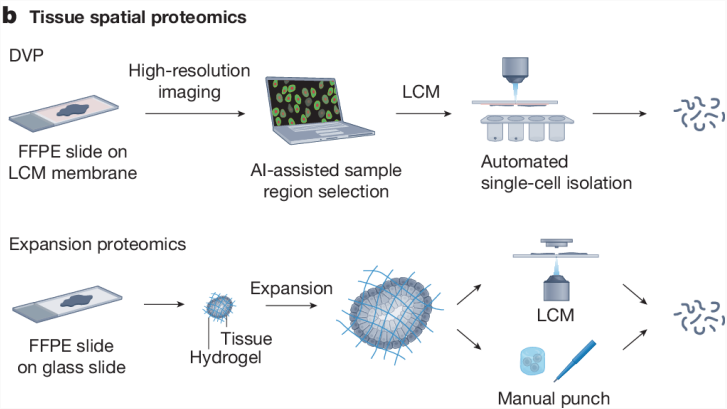
Examples summarized in the review

- Early SCoPE: about 1,000 proteins across 100 single cells.
- Label-free timsTOF: up to 2,000 proteins in individual HeLa cells.
- One-Tip: over 5,000 proteins in single oocytes.

These are different cell types and workflows, not a controlled ranking.

Guo et al., p. 905; refs. 111, 113, 116–118. Larger oocytes are not representative of typical small cells.

Spatial proteomics retains the tissue context



Guo et al., Fig. 2b (cropped from original). DVP: deep visual proteomics; LCM: laser-capture microdissection.

Deep visual proteomics: an evidence chain to treatment

1. Image tissue and classify cells or regions with AI-assisted analysis.
2. Microdissect the selected populations and measure their proteomes.
3. Interpret pathway differences in the original tissue context.

Example highlighted by the authors: toxic epidermal necrolysis
Spatial proteomics implicated JAK–STAT hyperactivation in this severe drug-induced skin disease. The cited study used the finding to support repurposing a JAK inhibitor.

Our interpretation: this is an informative translational example, not evidence that every proteomic signature yields a treatment or that all patients will respond.

Guo et al., p. 906; Mund et al., Nat. Biotechnol. (2022), ref. 119; Nordmann et al., Nature (2024), ref. 120.

Plasma profiling exposes the depth–scale tradeoff

Why it is difficult

- Plasma protein concentrations span over ten orders of magnitude.
- High-abundance proteins obscure lower-abundance signals.
- Depletion or enrichment adds depth, preparation complexity and potential variability.

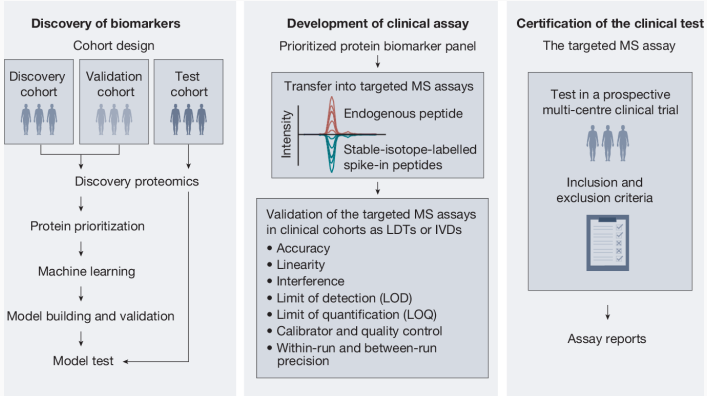
Review-era examples

- Neat plasma: routinely 300–500 proteins per injection.
- Astral: about 1,000 proteins, up to 100 samples/day.
- Perchloric-acid workflow: 1,600+ proteins with reduced cost.

These figures come from different workflows; depth alone does not establish clinical usefulness.

Guo et al., p. 906; refs. 125–133. Performance is sample- and protocol-dependent.

Discovery is only the first stage of a clinical test



Guo et al., Fig. 3 (original). LDT: laboratory-developed test; IVD: in vitro diagnostic.

What remains between a biomarker and routine care?

1. **Generalization:** adequately powered cohorts, independent validation and prospective multicentre testing.
2. **Analytical validity:** accuracy, precision, interference, linearity and detection/quantification limits.
3. **Transferability:** stable samples, internal standards and comparable measurements across instruments and laboratories.

The review's own qualification
Most tissue biomarker studies remain retrospective; few become clinically applicable tests.
Targeted assays with synthetic standards may transfer more readily than discovery workflows.

Guo et al., pp. 907–908 and Fig. 3. Clinical benefit must be demonstrated for the intended use.

AI: distinguish mature measurement tools from the outlook

Established roles described

- Predict peptide retention times and fragment spectra.
- Score identifications and extract DIA signals.
- Classify tissue images and support biomarker models.

Directions proposed by the authors

- Real-time acquisition decisions.
- Grounded LLM-assisted interpretation.
- Foundation models trained on perturbation proteomics.

These are a research agenda, not a common validated clinical platform.

Discussion point

Does a model learn biological state or the instrument, laboratory and cohort that produced the data?

Guo et al., pp. 903, 907–908 and Fig. 4. Discussion point is our interpretation.

How to read the review critically

1. **Coverage:** it organizes representative advances, but is not a systematic review or a head-to-head benchmark.
2. **Sensitivity:** more detected proteins does not guarantee accurate low-abundance quantification, PTM coverage or biological specificity.
3. **Translation:** promising cohort classification and mechanistic examples do not substitute for prospective clinical utility.

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Scope and clinical limitations: Guo et al., pp. 901, 906–908; disclosures p. 911. Critical framing is ours.

Take-home ideas and journal club questions

Take-home: improvements across preparation, separation, instrumentation and computation make direct protein measurements possible at smaller input and larger scale.

1. For our biological question, do we need abundance, occupancy, binding, activity or spatial context?
2. Which bottleneck dominates: sensitivity per cell, throughput per cohort, or reproducibility across sites?
3. What independent experiment would turn a proteomic association into a mechanistic or clinical claim?

Discussion synthesis of Guo et al. (2025). The review's optimism is a useful agenda; study design determines what a dataset can establish.

Sources and reading route

Primary review

Guo, T., Steen, J. A. & Mann, M. *Nature* **638**, 901–911 (2025).
`https://doi.org/10.1038/s41586-025-08584-0`

Read in this order

1. Fig. 1 and pp. 901–903: the measurement workflow.
2. Fig. 2 and pp. 904–906: functional, single-cell and spatial applications.
3. Fig. 3 and pp. 906–908: clinical assay development and remaining gaps.

All quantitative examples are attributed to the review's cited literature, not new measurements by the review authors. Figures 1–3 are reproduced from the article for journal club discussion.